$$\tau' = \beta\tau = \alpha^{3/2}\tau \tag{8}$$

and hence

$$\frac{\tau'^2}{l'^3} = \frac{\tau^2}{l^3}\,. \tag{9}$$

III.

Relative position is $\boldsymbol{r} = \boldsymbol{r}_\mathrm{B} - \boldsymbol{r}_\mathrm{A}$, with unit vector $\boldsymbol{e}_r = \boldsymbol{r}/r$. Particles are confined to their tubes; their net forces are the electrostatic components along their motion. Their equations of motion are

$$\begin{cases} m\boldsymbol{a}_\mathrm{A} = -\dfrac{kq^2}{r^2}\left(\boldsymbol{e}_r \cdot \boldsymbol{i}\right)\boldsymbol{i} \\ m\boldsymbol{a}_\mathrm{B} = \dfrac{kq^2}{r^2}\left(\boldsymbol{e}_r \cdot \boldsymbol{j}\right)\boldsymbol{j} \end{cases}, \tag{1}$$

Subtraction gives

$$m\boldsymbol{a} = \frac{kq^2}{r^2}\boldsymbol{e}_r\,, \tag{2}$$

Here $\boldsymbol{a} = \boldsymbol{a}_\mathrm{B} - \boldsymbol{a}_\mathrm{A}$ is the acceleration of B relative to A. By (2), B relative to A moves as a particle of mass m in a fixed-centre Coulomb field, with potential energy

$$U = \frac{kq^2}{r}\,. \tag{3}$$

Angular momentum and energy are conserved in a central field. Both refer to B relative to A; the initial conditions give

$$L = \left|-md\boldsymbol{i} \times \left(-u_0\boldsymbol{j} + u_0\boldsymbol{i}\right)\right| = md\sqrt{\frac{kq^2}{md}}\,, \tag{4}$$

$$E = \frac{1}{2}m\left(-u_0\boldsymbol{j} + u_0\boldsymbol{i}\right)^2 + \frac{kq^2}{d} = \frac{2kq^2}{d}\,. \tag{5}$$

The closest distance is r_m; there the velocity u_m is perpendicular to $r_\mathrm{m}\boldsymbol{e}_r$, giving

$$mr_\mathrm{m}u_\mathrm{m} = md\sqrt{\frac{kq^2}{md}}\,, \tag{6}$$

$$\frac{1}{2}mu_\mathrm{m}^2 + \frac{kq^2}{r_\mathrm{m}} = \frac{2kq^2}{d}\,. \tag{7}$$